



Machine Learning

Case Study 148: Network Intrusion Detection

1. Title

Network Intrusion Detection Using Machine Learning

2. Problem Statement

Modern networks generate large volumes of traffic. Detecting malicious network activity manually is difficult. An ML-based Intrusion Detection System (IDS) can analyze network traffic and identify normal and suspicious activities.

3. Objectives

- Analyze network traffic patterns.
- Identify characteristics of malicious traffic.
- Build an intrusion classification model.
- Compare different algorithms.
- Develop a prototype network intrusion detection system.

4. Machine Learning Algorithms

1. Logistic Regression
2. KNN
3. Decision Tree
4. Random Forest
5. Naive Bayes
6. Gradient Boosting

5. Comparative Study

Evaluate using:

- Accuracy
- Precision
- Recall
- F1-score



Machine Learning

- Confusion Matrix

For intrusion detection, students should particularly analyze **false negatives**.

6. Deployment

Create an application that accepts network-traffic features and predicts:

Normal / Intrusion

An advanced version can classify specific intrusion categories.

7. Final Analysis

Students should identify:

- Most important network traffic features.
- Best-performing algorithm.
- Most difficult attack categories.
- False-positive and false-negative patterns.
- Limitations of the IDS.

Questions to Be Answered Using ML

1. Can ML distinguish normal and malicious network traffic?
2. Which network features contribute most to intrusion detection?
3. Which algorithm provides the highest recall?
4. Which attack categories are difficult to identify?
5. How does class imbalance affect the model?
6. Can the model detect previously unseen traffic patterns?